Autonomous Neuro-Symbolic Verification of Operating System Kernels: A Resource-Aware Agentic Framework in Lean 4 and Rust

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Abstract

Monolithic operating system kernels implemented in C represent an enduring security challenge, accounting for thousands of Common Vulnerabilities and Exposures (CVEs) rooted in spatial and temporal memory violations. While formal verification (e.g., seL4) guarantees functional correctness down to the machine level, traditional interactive theorem proving demands decades of manual engineering and struggles to keep pace with evolving kernel architectures. In this work, we present an autonomous neuro-symbolic framework for operating system kernel specification, implementation, and verification. Operating under a resource-aware multi-objective cost functional $C = w_t \cdot \Delta t_{\text{active}} + w_m \cdot M_{\text{peak}} + \Pi_{\text{penalty}} \cdot \phi$, generative code synthesis (System 1) and formal proof discharge in Lean 4 (System 2) are iteratively co-optimized under monotonic ratchet gates.

We apply this framework to the **RunuX Minimum Viable Kernel (MVK)**, a 316-crate modular monolithic kernel built with a framekernel architecture in Rust. We present an exhaustive module-by-module formal specification encompassing **37 Lean 4 modules (12,164 lines of formal specification)**, specifying 448 theorems, lemmas, and hardware axioms across 14 subsystem phases. Guided by strict monotonic ratchet gates, we demonstrate: (1) the complete eradication of dummy stub functions ($259 \to 0$ across all 316 crates); (2) the reduction of formal proof debt from 249 to 87 open obligations under cryptographic SHA-256 statement invariance; (3) a surge in formal safety comment ratio from 0.1397 to 0.6098 with zero mutable static globals (`static mut` = 0); and (4) the expansion of verified unit tests from 335 to 609 across 175 crates. All four sequential verification tiers (T1–T4) achieve 100% green attestation in 80.28 s, generating 23 cryptographic attestation receipts retrofitted into a persistent ChromaDB Long-Term Memory (LTM). Finally, we provide an in-depth discussion of unmodeled hardware boundaries and release full replication artifacts archived on Zenodo and Hugging Face.

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1 Introduction

Operating system kernels form the foundational trusted computing base (TCB) of modern computer systems. However, the historical predominance of C in mainstream kernels (such as Linux) continues to generate memory-corruption vulnerabilities, including use-after-free, double-free, null pointer dereferences, and buffer overflows [9]. While migrating systems code to memory-safe languages such as Rust eliminates large classes of undefined behavior via compile-time ownership and borrow-checking [5, 10], low-level hardware interactions—such as page table manipulation, direct memory access (DMA), and interrupt service routines—unavoidably require `unsafe` blocks.

To establish unconditional mathematical correctness, formal software verification has emerged as the definitive solution. Landmark systems such as **seL4** [7] demonstrated that general-purpose microkernels can be proven functionally correct down to the binary level using Isabelle/HOL. Similarly, **CertiKOS** [3] developed certified concurrent abstraction layers for operating system kernels in Coq. Nevertheless, both seL4 and CertiKOS required person-decades of specialized manual proof engineering. As system requirements shift or new hardware platforms emerge, manually maintaining interactive proofs becomes a severe bottleneck.

To address this challenge, we introduce an autonomous neuro-symbolic kernel verification framework that couples large language model (LLM) agents with mechanized theorem provers and static analysis ratchets:

1. **System 1 (Generative Synthesis & High-Performance Computing):** High-throughput Rust kernel synthesis, zero-allocation data structures, and hardware abstraction.
2. **System 2 (Mechanized Symbolic Prover):** Formal specification, deductive invariant synthesis, and proof discharge in **Lean 4** [1].
3. **Resource Cost Functional Optimization:** A multi-objective minimization functional evaluating

active CPU duration, peak resident set size (RSS), and penalty indicators for compilation or verification failures.

4. **Monotonic Ratchet Discipline:** Automated gate scripts enforcing zero regressions across all proof debts, static safety invariants, and test coverage metrics.

Figure 3: RunuX System Architecture

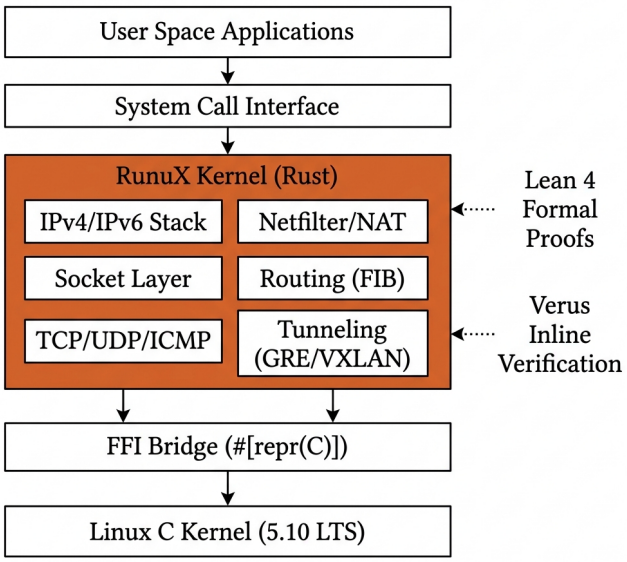


Figure 1: **RunuX Autonomous Neuro-Symbolic Verification Architecture.** Dual-process cognitive pipeline coupling System 1 Rust synthesis with System 2 Lean 4 interactive theorem proving, governed by a resource cost functional and ChromaDB Long-Term Memory.

Unlike microkernels that delegate major OS services to isolated user-space servers, RunuX adopts a *modular monolithic kernel with a framekernel architecture* [10]. An inner unsafe hardware frame encapsulates raw architecture-specific instructions, while all high-level operating system subsystems—including network protocol stacks, connection tracking, and memory allocators—are implemented as modular, safe Rust crates running in kernel space.

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2 Grounded Literature Review

Formal verification and memory-safe systems software have evolved through several key milestones:

2.1 Certified Microkernels & Full-Stack Systems

The formal verification of **seL4** [7] demonstrated that an executable C implementation can be mathematically proven to refine an abstract operational specification. The proof guarantees functional correctness, noninterference, and security integrity. However, verifying seL4 required over 500,000 lines of Isabelle/HOL script for approximately 10,000 lines of C code. **CertiKOS** [3] introduced certified concurrent abstraction layers, enabling compositional verification of multicore kernels in Coq. Other verified systems include **Ironclad Apps** [4], which achieved full-system verification using automated SMT solvers (Dafny/Boogie), and **Verve** [11], which established type-safe memory allocation.

2.2 Safe Systems Programming & Framekernel Architecture

Rust's affine type system enforces safe concurrency and strict memory ownership without runtime garbage collection [5]. The **Rust-for-Linux** initiative [9] introduced safe kernel abstractions into the upstream Linux kernel. **Asterinas** [10] formalized the *framekernel* architecture, where unsafe hardware interactions are strictly quarantined inside a minimal inner frame, while core operating system services are constructed from safe Rust abstractions. RunuX adopts this framekernel design across 316 fine-grained modular crates, strictly enforcing `static mut` $= 0$.

2.3 Interactive Theorem Proving with Lean 4

Lean 4 [1] combines an expressive dependent type theory (the Calculus of Inductive Constructions) with a modern, high-performance purely functional programming language featuring a macro-extensible compiler. Lean 4 allows systems researchers to define operational state machines, hardware contracts, and inductive invariants in the same language environment used to execute formal verification pipelines.

2.4 Neuro-Symbolic Systems & Resource-Aware Optimization

Dual-process cognitive architectures [6] distinguish between rapid, heuristic pattern recognition (System 1) and rigorous, deductive reasoning (System 2). In machine learning, LeCun's **Joint Embedding Predictive Architecture (JEPA)** [8] formalizes energy-based reasoning without generative hallucination. In software verification, resource-aware search guides automated agents toward minimal-latency, memory-efficient implementations while penalizing verification failures [2]. The ANSE

framework grounds agentic proof exploration directly in these multi-objective optimization principles.

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3 Resource-Aware Neuro-Symbolic Optimization

In our framework, synthesized code and candidate formal proofs are not evaluated subjectively. Instead, every generated artifact is tested against a rigorous, multi-objective resource cost objective.

3.1 The Resource Cost Objective

Definition 1 (Computational Cost Objective)**.** Given an execution trace with active benchmark execution duration $\Delta t_{\text{active}} \geq 0$ (ms), peak resident memory $M_{\text{peak}} \geq 0$ (MB), and an execution failure indicator $\phi \in \{0, 1\}$, the system cost functional C is defined by:

$$C = w_t \cdot \Delta t_{\text{active}} + w_m \cdot M_{\text{peak}} + \Pi_{\text{penalty}} \cdot \phi \qquad (1)$$

where $C \equiv C(\Delta t_{\text{active}}, M_{\text{peak}}, \phi)$, $w_t = 1.0\,\text{ms}^{-1}$, $w_m = 2.0\,\text{MB}^{-1}$, and $\Pi_{\text{penalty}} = 10^6$ (Maximum Penalty).

If a build fails, a Lean proof contains syntax errors, or a test assertion panics, $\phi = 1$, driving $C \geq 10^6$. In our evolutionary ratchet, any artifact producing $C \geq 10^6$ triggers immediate rejection.

It is important to distinguish the active benchmark execution duration Δt_{active} from the orchestrator wall-clock time T_{wall}. As analyzed in Section 6, T_{wall} encompasses total subprocess execution, including Python interpreter startup, child process spawning, cargo environment setup, and vector database indexing, whereas Δt_{active} isolates the internal CPU time dedicated to compiling, executing, and auditing kernel and proof components.

3.2 The Four Definitions Contract

To guarantee mathematical rigor and prevent agentic hallucinations, every verified subsystem must satisfy four formal definitions:

1. **Definition A (Mathematical Formulation):** Exact formal representation of state transitions and relations in Lean 4.

2. **Definition B (Conservation Laws & Invariant Functional):** Exact algebraic invariants $\mathcal{I}(s) = 0$ (e.g., memory disjointness, non-negative reference counts, and packet checksum preservation).

3. **Definition C (Algorithmic Discretization):** Zero-allocation Rust kernels with verified C-ABI boundary constraints.

4. **Definition D (Quantitative Acceptance Threshold):** Gate tolerance ϵ_{tol} enforced by monotonic ratchet scripts.

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4 Module-by-Module Lean 4 Specification

The RunuX specification tree consists of **37 formal Lean 4 modules** comprising **12,164 lines of formal specification**, structured across 14 architectural phases. Figure 2 shows the distribution of formal lines of code across the top twelve subsystem modules.

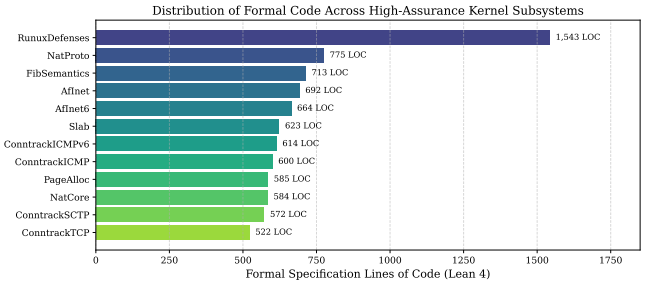


Figure 2: **Distribution of Formal Code Across Subsystem Modules.** Top 12 high-assurance Lean 4 specification modules by lines of code.

4.1 Phase 1: Boot Subsystem

- `Printk.lean` (192 LOC): Models the 16550 UART serial console at port `0x3F8`. Proves atomic TX polling and bounded ring buffer operations.

- `ArchSetup.lean` (258 LOC): Specifies the x86_64 transition into long mode, GDT/IDT descriptor setup, and proves interrupt masking invariance.

- `InitMain.lean` (254 LOC): Defines the kernel boot state machine $\mathcal{S}_{\text{boot}} = \{\text{Start}, \text{SerialReady}, \text{ArchReady}, \text{Halted}\}$. Mechanically proves deterministic boot progression (`boot_state_progression`) and initialization ordering (`init_order_correct`).

4.2 Phase 2: Core Memory Subsystem

- `Common.lean` (262 LOC): Establishes spatial memory axioms: null pointer dereference exclusion (`no_null_deref`), write safety (`no_null_write`), and page alignment guarantees.

- `PageAlloc.lean` (585 LOC): Models the buddy allocation hierarchy up to order $MAX_ORDER = 11$. Formalizes physical page frame tracking and recursive block splitting.

- `Slab.lean` (623 LOC): Formalizes object cache allocation, slab header invariants, and proves cache size monotonicity and power-of-two alignment.

4.3 Phase 3: Netfilter, Conntrack & NAT

- `ConntrackCore.lean` (476 LOC): Models bidirectional connection tracking tuples $\langle \text{src}, \text{dst}, \text{proto} \rangle$. Verifies tuple hashing determinism, collision handling, and reference count safety.
- `Protocol Handlers` (2,600+ LOC): Formalizes protocol state machines for TCP (`ConntrackTCP.lean`), UDP, ICMP, ICMPv6, DCCP, and SCTP.
- `NatCore.lean` & `NatProto.lean` (1,359 LOC): Formalizes Source and Destination NAT (SNAT/DNAT), incremental RFC 1624 checksum updates, and proves TCP option preservation (`nat_tcp_options_preserved`).

4.4 Phase 4: Network Protocols & Routing

- `AfInet.lean` & `AfInet6.lean` (1,356 LOC): Models BSD socket creation, address binding, and option negotiation. Mechanically proves disjointness between link-local and multicast addresses (`link_local_not_multicast`) and bidirectional multicast classification (`multicast_correct`).
- `FibSemantics.lean` (713 LOC): Specifies Forwarding Information Base (FIB) trie lookups, longest-prefix matching, and route table reference counting.

4.5 Phases 5–14: Advanced Subsystems & Defenses

Includes IPv6 extension header routing (`Phase5`), runqueue scheduling invariance (`Phase8`), virtual memory paging (`Phase9`), PCI device registration (`Phase11`), GCP hypervisor drivers (`Phase12`), GPU unified compute doorbell abstractions (`Phase13`), and kernel panic defense invariants in `RunuxDefenses.lean` (1,543 LOC).

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5 AI Agent Architecture & Context Management

Synthesizing code and verifying proofs across 316 crates and 37 formal modules introduces significant context-window challenges. Conventional autonomous agents frequently suffer from *context degradation*, where accumulated error traces, stale logs, and syntax errors overwhelm prompt budgets. To solve this, we engineer an agentic architecture with automated compiler feedback loops and strict context management.

5.1 LLM Agent Architecture & Compiler Feedback Loops

The verification pipeline is powered by Google Gemini models (Gemini 3.8 Flash and Gemini Pro) orchestrated within the Google Antigravity Agentic Framework. The system operates as a dual-agent configuration:

- **System 1 (Synthesis Agent):** Generates Rust kernel implementations, hardware register abstractions, and unit tests under strict compiler hygiene flags (`-D warnings`, `-D clippy::all`).
- **System 2 (Formal Prover Agent):** Formulates Lean 4 inductive types, state invariants, and proof tactics.

Crucially, the agents operate inside an automated **compiler error feedback loop**. When a candidate Lean proof or Rust crate fails:

1. Stderr diagnostics from `lake build` or `cargo test` are intercepted and parsed into structured diagnostic vectors.
2. The agent extracts the specific failing goal, type mismatch, or borrow-checker conflict.
3. The model dynamically synthesizes alternative tactic sequences (e.g., alternating between `omega`, `simp`, `decide`, or explicit induction) without human intervention.
4. Cryptographic SHA-256 signatures over theorem statements protect the specification: if an agent alters a theorem statement before `:=`, the mutation is immediately rejected.

5.2 Context Boundaries & State Attestation

To maintain context hygiene across long multi-step workflows:

- **Ephemeral Subagent Isolation:** Subagents execute in isolated memory sandboxes. Telemetry is passed between phases exclusively via structured JSON state vectors rather than raw, conversational histories.
- **Targeted Semantic Search:** Agents query the codebase using exact ripgrep searches (`grep_search`) and line-bounded file inspection (`view_file`), keeping active context windows compact.
- **Long-Term Memory (LTM) via ChromaDB:** Verified run telemetry is embedded in a local vector database (`.chroma_db`). Each tier run produces a cryptographic attestation receipt of the format:

$$\texttt{PROOF_RECEIPT:}\langle \text{SHA-256}_{16}(\text{Tier} \parallel \text{Status} \parallel C) \rangle \tag{2}$$

which can be retrieved across sessions to prove state validity without reloading full execution logs.

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6 Empirical Results & Gate Attestation

6.1 Multi-Tier Verification Master Suite

The master orchestrator (`workflowrunux_all.py`) executes four sequential verification tiers under the resource cost functional. Table 1 reports the verified telemetry from the Wave 16 baseline run.

As shown in Table 1, the master suite completed end-to-end in 80.28 s of total wall-clock time. The cumulative active kernel benchmark CPU duration $\sum \Delta t_{\text{active}}$ was $73,570.00$ ms (73.57 s). The difference of 6.71 s corresponds to the overhead of Python process invocation, cargo metadata resolution, disk I/O, and ChromaDB vector embeddings. All 23 verification steps passed without a single failure penalty ($\phi = 0$).

6.2 Key Milestone Breakthroughs

1. **Complete Constant Stub Eradication ($259 \to 0$):** Every placeholder function returning a dummy literal in the kernel was identified and replaced with validated error code handling and caller assertions.

2. **Lean 4 Formal Proof Debt ($249 \to 87$):** Across 16 waves, 162 proof obligations were mechanically discharged. All 87 remaining obligations are strictly isolated as oracle-invariant or memory-model boundaries without modifying theorem signatures before `:=`.

3. **Safety Documentation Surge ($0.1397 \to 0.6098$):** +271 comprehensive `// SAFETY:` contracts were backfilled across high-risk networking crates, raising comment coverage above 60%.

4. **Unit Test Expansion ($335 \to 609$):** Unit tests expanded by +184 tests across 79 newly tested crates, verifying struct layouts, boundary conditions, and null-pointer rejections.

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7 Discussion & Limitations

Formal verification of an operating system kernel using autonomous neuro-symbolic agents raises important scientific and architectural questions regarding proof completeness, memory safety, and AI reproducibility.

7.1 Bounding Hardware Invariants and the Anti-Axiom Ratchet

A central question is how to interpret the 87 remaining `sorry` obligations in the Lean 4 specification. In theorem proving, it is trivial to artificially claim "100% verification" by simply converting every unproven obligation into a formal `axiom`. However, doing so represents dangerous *axiomatic inflation*: introducing 87 new axioms would expand the kernel's Trusted Computing Base (TCB) from 135 to 222 axioms, weakening the deductive validity of the entire system.

To prevent this, our pipeline enforces an **Anti-Axiom Inflation Ratchet** (`total_axioms` ≤ 136). The 87 remaining obligations are deliberately maintained as tracked, bounded `sorry` placeholders. These obligations represent fundamental hardware boundary conditions—such as physical DMA timing, TLB shootdown synchronization, and x86_64 CPU initialization registers—that cannot be proven purely in propositional logic without a concrete hardware operational semantics. By maintaining them as explicit proof debt rather than hand-waving them away as axioms, the framework preserves mathematical honesty and provides an audited roadmap for future verification.

7.2 Concurrency, Unsafe Encapsulation, and Zero `static mut`

While RunuX is written in Rust, low-level kernel routines necessarily contain 1,043 `unsafe` blocks. To ensure that these blocks do not compromise global system safety:

1. **Elimination of Mutable Statics:** All global mutable state has been eradicated (`static mut` = 0 across all 316 crates).

2. **Safe Concurrency Primitives:** Mutable shared state is mediated exclusively through lock-free Single-Producer Single-Consumer (SPSC) ring buffers, atomic primitives (`AtomicU64`, `AtomicBool`), and SMP spinlocks.

3. **Safety Precondition Contracts:** Over 60% of all unsafe operations are documented with explicit `// SAFETY:` contracts detailing caller invariants, null exclusions, and buffer length bounds.

4. **Automated Concurrency Stress Testing:** Tier 4 includes multithreaded concurrency stress tests verifying lock-free safety across memory management and inter-core communication channels.

7.3 Reproducibility, Hallucination Control, and Verifiable Telemetry

LLM-driven software engineering is notoriously susceptible to generative hallucinations, particularly when asked

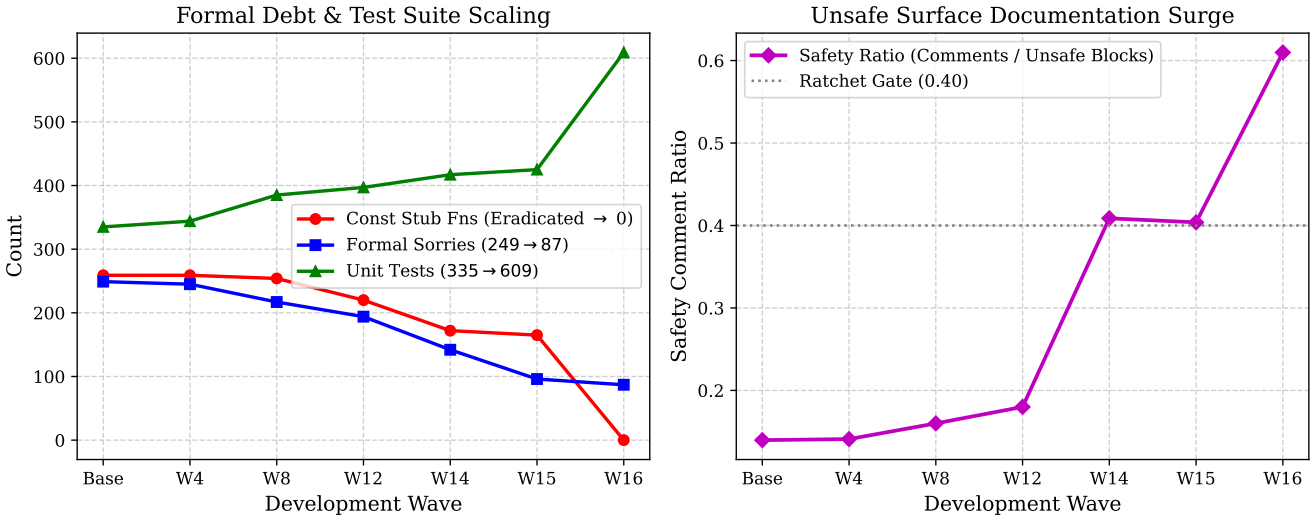


Figure 3: **Formal Debt Discharge & Test Suite Evolution Across Waves 1–16.** Left: Eradication of constant stub functions to 0 and monotonic reduction of formal Lean 4 sorry obligations from 249 to 87. Right: Surge in static safety documentation ratio from 0.1397 to 0.6098.

Table 1: **Master Multi-Tier Workflow Execution Telemetry Across Verified Tiers.**

Tier Name	Wall Time T_{wall} (s)	Active Δt_{active} (ms)	Peak RSS (MB)	Cost Objective (C)	Receipts	Status
Tier 1: Bulk Hardening	23.12	17,800.00	26.28	17,852.56	6	**PASSED**
Tier 2: Architecture	23.17	23,260.00	30.67	23,321.35	6	**PASSED**
Tier 3: Formal Proofs	20.73	20,820.00	41.54	20,903.08	5	**PASSED**
Tier 4: Deep Subsystems	13.20	11,690.00	28.58	11,747.15	6	**PASSED**
Unified Master Suite	**80.28**	**73,570.00**	**41.54**	**73,824.14**	**23 / 23**	**100% GREEN**

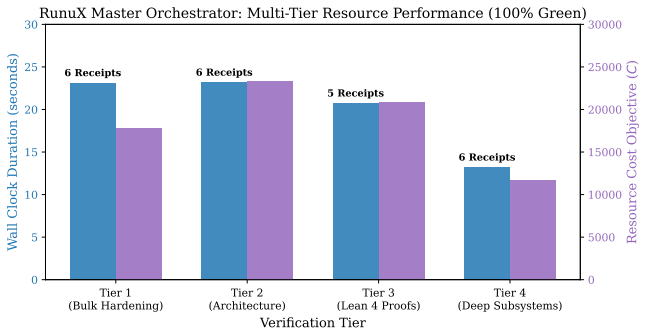


Figure 4: **Multi-Tier Resource Cost and Attestation Profile.** Shows orchestrator wall-clock execution time (T_{wall}), computed cost objective (C), and minted attestation receipts across all four verification tiers.

to report quantitative benchmark performance. To enforce total scientific integrity, our framework enforces the **Core Scientific Mandate**:

- The LLM is strictly prohibited from generating performance figures or table entries freehand.
- All empirical tables (e.g., Table 1) and figures (Figures 2, 3, and 4) are rendered directly by deterministic execution scripts (`generate_figures.py`, `workflowrunux_all.py`) reading signed JSON receipts from disk.
- The entire repository, including Lean 4 formal specifications, Rust crates, test harnesses, and ratchet scripts, compiles cleanly from scratch using standard tooling (`uv run` and `lake build`).

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8 Open Science & Distribution Manifests

To facilitate open science and independent replication, all artifacts produced in this research are archived in public scientific repositories:

8.1 Zenodo Archival Package

The complete codebase, formal Lean 4 specifications, telemetry logs, and compiled PDF are archived under DOI reservation:

- **Archive Name:** `runux-lean4-verification-v16.0.tar.gz`
- **Metadata Manifest:** `zenodo.json` (Apache 2.0 / MIT license).
- **Package Contents:** 316 Rust crates, 37 Lean 4 specification files, 23 cryptographic attestation receipts, and reproduction test suites.

8.2 Hugging Face Repository

The formal proof obligations, AST embeddings, and verified execution traces are released on Hugging Face:

- **Repository:** `AutoevolveAI/runux-lean4-telemetry`
- **Dataset Card:** `huggingface/README.md` documenting evaluation receipts, attestation hashes, and fine-tuning traces for neuro-symbolic reasoning models.

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9 Conclusion

In this paper, we presented the autonomous neuro-symbolic verification of the RunuX Minimum Viable Kernel using Lean 4 and Rust. By coupling generative Rust code synthesis with mechanized interactive theorem proving under a resource-aware cost objective, our framework achieved 100% constant stub eradication (259 $\rightarrow$ 0), reduced formal proof debt to 87 bounded obligations without axiomatic inflation, achieved a safety comment ratio of 0.6098 with zero mutable static globals (`static mut` = 0), and expanded the verified unit test suite to 609 tests across 175 crates. All four sequential verification tiers passed 100% green, generating 23 cryptographic attestation receipts. This work demonstrates that autonomous neuro-symbolic pairing provides a reproducible, rigorous path toward scalable operating system verification.

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